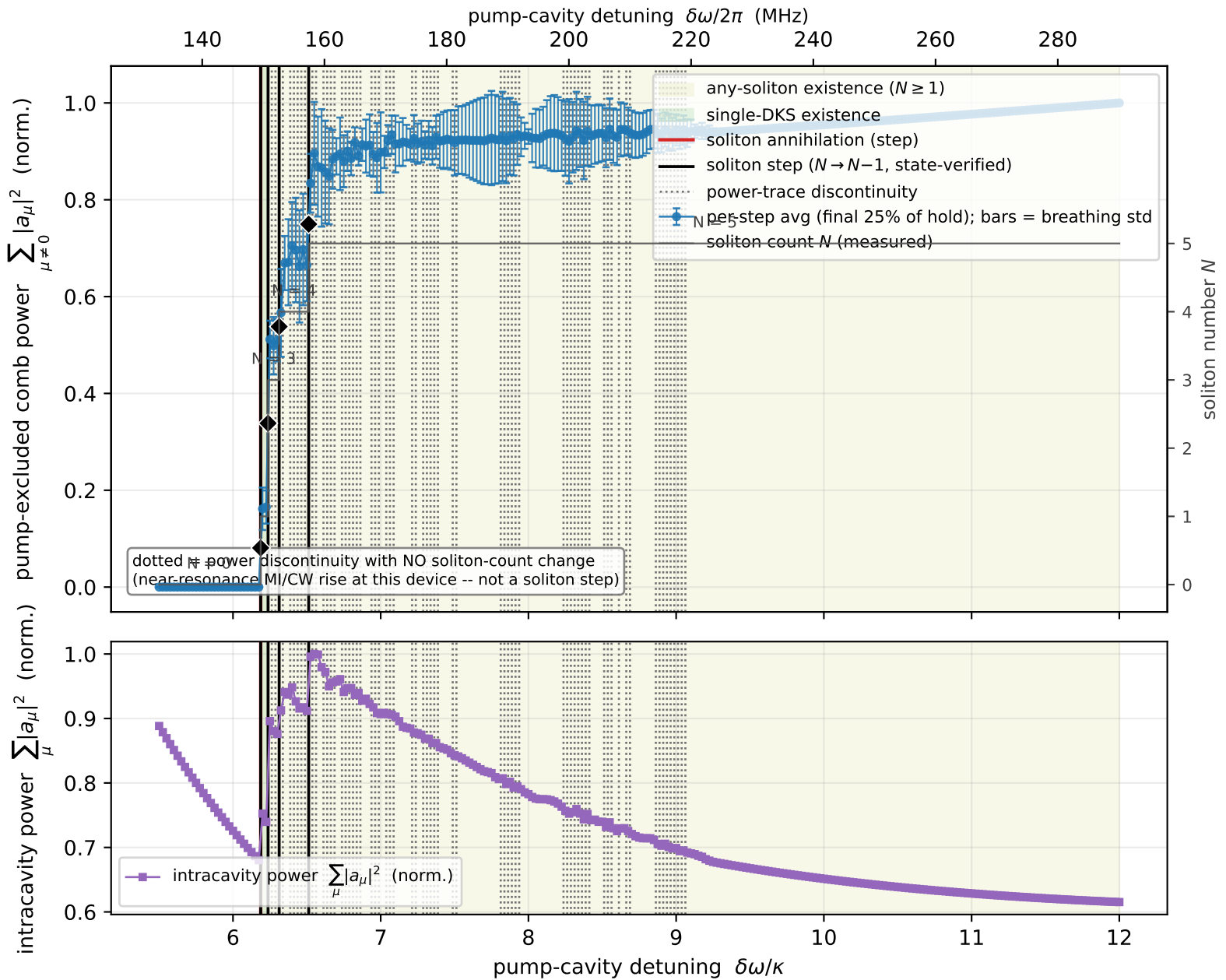




5-soliton staircase, $p_{in} = 0.214$ W, $n_{\tau} = 4096$. Deterministic symmetry-broken seed (position_seed=1, jitter=0.25) at 12κ , warm continuation swept DOWN, hold 2000 RT/step, averaged over the final 25% (cycle-averages the breather). Thermo-optic model ON (deterministic, noise off). The solitons annihilate sequentially at the branch's lower edge -- the staircase; the final $1 \rightarrow 0$ annihilation must never be FORCED to register as a power step (matched only if the untouched detector resolves it naturally on the plotted trace). Green = single-DKS existence region. Olive = any-soliton ($N \geq 1$) region. Solid black lines mark STATE-VERIFIED soliton steps (power discontinuity coinciding with a measured soliton_count transition); dotted lines mark power discontinuities without a state change. Thin gray staircase = measured soliton count N (twin axis). Raw data, no smoothing.